EXPERIMENT: LONGEST SUBSTRING WITHOUT REPEATING CHARACTERS

Experiment No.: 06

Title: Find the Longest Substring Without Repeating Characters

1. Objective

To write a Java program to find the longest substring without repeating characters and display its length and the substring.

2. System Requirements

Hardware: Computer/Laptop
Software: JDK (Java Development Kit), any Java-supported IDE or command prompt.

3. Algorithm / Logic

1. Read a string from the user.
2. Use two pointers to keep track of the current substring.
3. Check each character for repetition in the current substring.
4. If a repeated character is found, move the starting position.
5. Calculate the current substring length.
6. Store the maximum length and starting position.
7. Extract and display the longest substring and its length.

4. Logic / Flow

Start
↓
Read input string
↓
Check characters for repetition
↓
Update start position if repeated
↓
Calculate current length
↓
Store maximum substring
↓
Display longest substring and length
↓
Stop

5. Source Code

```
import java.util.Scanner;

public class LongestSubstring {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String s = sc.nextLine();

        int start = 0;
        int maxLength = 0;
        int bestStart = 0;

        for (int end = 0; end < s.length(); end++) {

            for (int j = start; j < end; j++) {

                if (s.charAt(j) == s.charAt(end)) {
                    start = j + 1;
                    break;
                }
            }

            int currentLength = end - start + 1;

            if (currentLength > maxLength) {
                maxLength = currentLength;
                bestStart = start;
            }
        }

        String longest =
                s.substring(bestStart, bestStart + maxLength);

        System.out.println("Length of longest substring: " + maxLength);
        System.out.println("Longest substring: " + longest);

        sc.close();
    }
}
```

6. Compilation and Execution

```
javac LongestSubstring.java
java LongestSubstring
```

7. Sample Output

```
Enter a string: abcabcbb
Length of longest substring: 3
Longest substring: abc
```

8. Result

The Java program was successfully implemented to find and display the longest substring without repeating characters.